BERT-Based Fine-Tuning for Automated Tagging of Robbery Crime Narratives

Lenin G. Falconi[1 [0000−0003−4402−6643]],
Myriam Hernandez-Alvarez[1 [0000−0003−4718−0400]], and
Ángel Leonardo Valdivieso Caraguay[1 [0000−0002−3502−020X]]

Escuela Politécnica Nacional, Quito 170525, Ecuador
https://www.epn.edu.ec
{lenin.falconi,myriam.hernandez,angel.valdivieso}@epn.edu.ec

Abstract. Accurate classification of crime narratives is vital for reliable public safety statistics. In Ecuador, the Comisión Especial de Estadística de Seguridad, Justicia, Crimen y Transparencia (CEESJCT) manually categorizes robbery incident reports, which is a time-consuming process. While transformer-based models have shown success in natural language processing tasks, their application to Ecuadorian legal and security texts in the Spanish language remains underexplored. This study addresses this gap by developing an automated classification system using a BERT model tailored to Spanish robbery narratives. Utilizing transfer learning and subsequent fine-tuning on an expanded, labeled dataset, the system significantly improves classification performance. Initial transfer learning achieved moderate accuracy (80.5%) but faced difficulties with semantically similar categories. Fine-tuning notably increased minority-class recall (up to 30%) with an improved accuracy (90.3%). The final implementation, which increased the number of categories to 11, achieved 95.5% accuracy with robust and consistent results on both police and judicial narratives. Collaboration with Ecuadorian institutions, including the Fiscalía General del Estado (FGE) and Instituto Nacional de Estadística y Censos (INEC), ensures model credibility.

Keywords: Legal Natural Language Processing · NLP · Natural Language Processing · transformers · fine-tuning · transfer learning.

1 Introduction

Deep learning (DL) has achieved remarkable success in text classification (TC), often outperforming traditional Machine Learning (ML) methods[14]. However, DL models typically require vast amounts of data and computational resources. For specialized domains with limited data, such as legal text processing, *Transfer Learning (TL)* and *Fine-Tuning (FT)* offer a viable path to adapt large pre-trained models to niche tasks [28,9].

The legal domain poses distinct challenges for Natural Language Processing (NLP), characterized by complex terminology, lengthy documents, and intricate structural features. These factors have fostered the emergence of Legal NLP (LNLP) as a specialized subfield [4,29]. Although there has been increasing interest in judicial outcome prediction [15,11,26], legal article prediction from text, and document classification [27,6], available benchmark datasets such as LexGlue primarily target English, thereby neglecting other languages. While transformer models such as mBERT have facilitated cross-lingual transfer, the unique characteristics of legal terminologies and system-level differences underscore the need for domain-specific pretraining (e.g., LegalBERT [5]) or hybrid approaches [13], especially in the context of low-resource languages [20].

In Ecuador, the national penal code (COIP) defines robbery broadly, but for public security analytics, it must be sub-classified into types like home, street, or vehicle robbery. This classification is manually performed by the CEESJCT from crime incident reports. To automate this, we propose a classifier based on the distilbert-base-multilingual-cased transformer model [19], which is well-suited for multilingual applications. Our development involved three phases: (1) Initial TL using the transformer as a feature extractor, which yielded 80.5% accuracy; (2) Full model FT, which improved accuracy to 90.3%; and (3) a final FT phase on a large-scale dataset of 1.1 million narratives, expanding from 6 to 11 categories and achieving a final accuracy of 95.5%.

This research makes three key contributions: (1) a novel, large-scale dataset of over one million annotated Spanish-language robbery narratives from Ecuadorian police reports; (2) an effective three-phase FT approach that adapts a pre-trained BERT model to the Ecuadorian legal context, achieving 95.5% accuracy; and (3) the successful deployment of our model in Ecuador's FGE, automating a critical manual classification process. This represents, to our knowledge, the first application of transformer models for crime narrative analysis in Ecuadorian Spanish.

The remainder of this article is structured as follows: Related works are presented in Section 2. The proposed methodology, including dataset generation, is detailed in Section 3. Section 4 presents the experimental results. Finally, Section 5 discusses the main findings, conclusions, and directions for future work.

2 Related Works

Legal Text Classification (LTC) refers to the task of categorizing legal documents into predefined juridical categories. Research has increasingly adopted transformer architectures, which use a self-attention mechanism to capture long-range dependencies in text more effectively than prior architectures [24,23] and are considered state-of-the-art in NLP tasks. BERT [7], a notable application of the Transformer encoder, generates deeply bidirectional contextual representations through an exclusively encoder-based architecture.

A review of LTC research, as summarized in Table 1, reveals a predominant focus on English corpora, whereas the present work constitutes a novel application to Spanish legal documents from Ecuador. Transformer architectures, notably BERT, have been shown to outperform LSTM-based models on large, multi-label legal datasets by employing strategies such as gradual layer unfreezing and domain-specific pretraining, as demonstrated in prior studies [21]. Furthermore, research has extended these findings by exploring zero-shot cross-lingual transfer; for instance, models fine-tuned exclusively on English data have achieved strong performance on French and German texts [22]. Additional validation of transformer superiority over traditional machine learning methods, even when limited data is available, has been provided for Turkish legal texts [2]. Moreover, domain-adapted transformer models, including Legal-LongFormer and Legal-BERT, have demonstrated superior performance compared to their general-domain counterparts when applied to lengthy legal documents, such as U.S. Supreme Court decisions [25].

Table 1. Comparative analysis of legal text classification approaches

Ref	Model	Task	Dataset	Language	Key Results
[21]	BERT, RoBERTa, DistilBERT, XLNet, M-BERT	Multi-label classification	JRC-Acquis, EURLEX57K	Multilingual, English	0.661 (F1) on JRC-Acquis
[22]	M-BERT, M-DistilBERT	Cross-lingual transfer	EURLEX57K Extended	Models pre-trained in Multilingual. FT in English	34% improvement on French, 87% improvement on German
[2]	Transformer-based	Document classification	Turkish Legal Corpus	Turkish	DL models improve over traditional ML
[25]	BERT, RoBERTa, Legal-BERT, LongFormer, Legal-LongFormer	Long document classification	SCDB	English	80.1% accuracy (15 categories), 60.9% accuracy (279 categories) with Legal-BERT
This work	DistilBERT	Multi-label classification	CEESJCT robbery reports	Spanish	95.5% accuracy (11 categories)

The reviewed literature indicates that BERT models are widely utilized for LTC. Transformer-based architectures consistently outperform traditional ML methods in this domain [2]. FT pre-trained BERT's weights to the target task is a standard approach. Also, our methodology is comparable to that of [25], wherein models are trained for both broader and fine-grained legal categories. A recurrent challenge with BERT-based approaches concerns the processing of lengthy documents, due to BERT's tokenization limits.

Distinct from previous works, our study specifically addresses the classification of robbery types. Most literature focuses on broader legal categories, and while some studies may implicitly include robbery within categories such as "criminal law" or "property offenses", explicit attention to robbery classification is absent, marking our focus as potentially novel and expanding the exploration in the application of transformer-based models in the legal domain.

An additional distinguishing aspect of this study lies in the nature of its data. Our dataset consists of narratives drawn from citizen crime reports, police accounts, and prosecutor's office documents. As a result, it features a rich combination of technical language and non-technical descriptions provided by victims. The deployment of the model at the FGE, utilizing the proposed methodology, demonstrates significant potential for generating innovative solutions within the legal domain of governmental institutions.

3 Methodology

This section details the development of a transformer-based robbery classification model. The process involved three stages: dataset generation, initial TL on a 6-category dataset ($D_{\kappa=6}$) followed by FT, and concluding with FT on an expanded 11-category dataset ($D_{\kappa=11}$).

3.1 Dataset Generation

The initial target dataset, $D^T_{\kappa=6,N}$, comprises six crime categories from 671,146 robbery records provided by CEESJCT. After standard text pre-processing, including the removal of non-alphanumeric characters and conversion to lowercase, we analyzed statistical characteristics of the narrative length. To comply with DistilBERT's[19]) 512-token limit, we filtered narratives to be between 35 and 300 words, resulting in a dataset of $N = 431{,}669$ samples. This was stratified into training (63.32%), validation (15.83%), and test (20.85%) subsets.

Extended Taxonomy ($D_{\kappa=11}$) Dataset We constructed a larger dataset of 11 categories by integrating police reports with prosecutor's office records from 2014-2022. This process involved extracting, filtering, and merging narratives, followed by quality control, including text length filtering (50-400 words) and label standardization. The final dataset contained 1,109,335 records, partitioned into training (807,468 samples, approximately 72%), validation (201,867 samples, approximately 18%), and test (100,000 samples, approximately 10%) sets.

Table 2. Extended Taxonomy Dataset

Basic Taxonomy	Extended Labels	Total
HOME ROBBERY	HOME ROBBERY	172,264
STREET ROBBERY	STREET ROBBERY	421,497
BUSINESS ROBBERY	BUSINESS ROBBERY	74,088
VEHICLE PARTS AND ACCESSORIES THEFT	VEHICLE PARTS AND ACCESSORIES THEFT	154,546
CAR THEFT	CAR THEFT	90,038
MOTORCYCLE THEFT	MOTORCYCLE THEFT	119,128
NO INFORMATION	OTHER ROBBERIES	43,468
	WATER VESSEL ROBBERY	9,407
	SOCIAL ORGANIZATION ROBBERY	3,087
	EDUCATIONAL INSTITUTION ROBBERY	17,252
	PUBLIC INSTITUTION ROBBERY	4,560
Total		1,109,335

3.2 Model Training

Problem Formulation We address the problem of classifying crime report narratives of robbery into k predefined categories. Since a language model estimates the conditional probability of the next token given a preceding sequence of tokens, TC can be formalized as shown in (1). Training a TC model thus implies learning a function f that maps a target dataset D^T, consisting of text sequences $X_i \in D = \{X_1, X_2, \ldots, X_N\}$ to one of k categories ((2)).

For this work, we selected the distilbert-base-multilingual-casedmodel [19], a distilled version of BERT, which retains 97% of BERT's language understanding capabilities while significantly reducing the number of parameters. Moreover, it supports multilingual tasks, including Spanish, aligning well with the linguistic requirements of our dataset.

$$\hat{y} = \underset{y \in \{1,\ldots,k\}}{\operatorname{argmax}} P(y \mid w_1, w_2, \ldots, w_i) \tag{1}$$

$$f : D \longrightarrow \mathbb{Z}^k \tag{2}$$

To enable processing by a transformer-based language model, textual sequences must first undergo tokenization and be transformed into numerical representations. This involves segmenting the text into discrete tokens and subsequently mapping these tokens to vector embeddings. Let Tokenize : $\mathcal{X} \rightarrow \mathbb{Z}^\lambda$ be the tokenization function that maps text sequences to token IDs, where $\mathcal{X}$ is the text space and λ is the maximum sequence length. For each input sequence $X_i \in D_N^T$, the tokenizer produces:

$$(\boldsymbol{\nu}_i, \boldsymbol{\tau}_i) = \text{Tokenize}(X_i) \tag{3}$$

where:

- $\boldsymbol{\nu}_i \in \mathbb{Z}^\lambda$ are the token IDs
- $\boldsymbol{\tau}_i \in \{0, 1\}^\lambda$ is the attention mask
- $\lambda = 512$ for distilbert-base-multilingual-cased

The token IDs are passed through the embedding layer, which maps each token to a $d = 768$ dimensional vector representation. The embedding matrix $W_{embedding} \in \mathbb{R}^{119{,}547 \times 768}$ uses a $V = 119{,}547$ vocabulary, where each row corresponds to a unique token, thereby providing dense vector representations.

$$\mathbf{E} = \text{Embedding}(\boldsymbol{\nu}_i) \in \mathbb{R}^{\lambda \times d} \tag{4}$$

The encoded numerical representations of the text are then processed by distilbert-base-multilingual-cased:

$$\mathbf{H} = \text{BERT}(\mathbf{E}, \boldsymbol{\tau}_i) \in \mathbb{R}^{\lambda \times d} \tag{5}$$

The classification layer comprises a feedforward layer that maps the hidden state vector output from the encoder (or CLS embedding) $\mathbf{h}_{[\text{CLS}]}$ to the k categories:

$$\hat{\boldsymbol{y}}_i = \mathrm{softmax}(\mathbf{W}\mathbf{h}_{[\mathrm{CLS}]} + \mathbf{b}) \in \mathbb{R}^k \tag{6}$$

where:

- $\mathbf{h}_{[\mathrm{CLS}]} \in \mathbb{R}^d$ is the input feature vector of $d = 768$
- $\mathbf{W} \in \mathbb{R}^{k \times d}$ is the classification weight matrix, mapping from embedding space to k classes.
- $\mathbf{b} \in \mathbb{R}^k$ is the bias vector
- k is the number of target categories
- A softmax activation to obtain probabilities $p_i = \frac{\exp(y_i)}{\sum_{j=1}^{k} \exp(y_j)}$, where p_i is the probability of class i

In this work, we experiment with TL and FT to adapt the weights of a pre-trained model from a large source dataset D^S to the characteristics of a target dataset D^T. We adopt the TL definition provided by [8]:

Definition 1 (Transfer Learning). *Given an original domain $\mathcal{D}^S$ with an original learning task $\mathcal{T}^S$, a target domain $\mathcal{D}^T$, with a target task $\mathcal{T}^T$, the TL operation seeks to improve the learning of a prediction function $\phi^{\mathrm{T}}(\cdot)$ in $\mathcal{D}^T$ using the knowledge acquired in $\phi^{\mathrm{S}}(\cdot)$ in $\mathcal{T}^S$, through ϕ^S; where $\mathcal{D}^S \neq \mathcal{D}^T$ or $\mathcal{T}^S \neq \mathcal{T}^T$.*

Considering the mentioned definition of TL, (7) represents the TL operation applied to the text classification problem, starting from a pre-trained natural language model; where L is the number of layers of the pre-trained model.

$$\mathbb{T}_{\mathbb{L}} \left\langle \phi^{\mathrm{S}}(\mathcal{D}^S), \mathcal{D}^T \right\rangle = \mathcal{A}\left(\phi^{\mathrm{S}}(\mathcal{D}^T)\Big|_0^L \right) \tag{7}$$

Consequently, the prediction generation is represented by Equation (8)

$$\mathcal{Y}^T = \mathbb{T}_{\mathbb{L}} \left\langle \phi^{\mathrm{S}}(D^S), \mathcal{D}^T \right\rangle = \phi^{\mathrm{T}}(D^T) \tag{8}$$

Similarly, we use the definition of FT as presented in [8]:

Definition 2 (Fine Tuning). *Given an original domain $\mathcal{D}^S$ with an original learning task $\mathcal{T}^S$, a target domain $\mathcal{D}^T$, with a target task $\mathcal{T}^T$, the FT procedure enhances the learning of a target function $\mathcal{Y}^T = \phi^{\mathrm{T}}(\mathcal{I}^T)$ by retraining r layers of the original model $\mathcal{Y}^S = \phi^{\mathrm{S}}(\mathcal{I}^S)$ with L layers and $\gamma = L - r$.*

The definition for FT is provided in (9), whereas the special case $\gamma = 0$, where $L = r$, is given in (10) and corresponds to full model tuning.

$$\mathbb{F}_{\mathbb{T}} \left\langle \phi^{\mathrm{S}}(\mathcal{D}^S), \mathcal{D}^T \right\rangle = \mathcal{A}\left(\phi^{\mathrm{S}}\left(\phi^{\mathrm{S}}(\mathcal{D}^T)\Big|_0^{\gamma-1} \right)\Big|_{\gamma}^{L} \right) \tag{9}$$

$$\mathbb{F}_{\mathbb{T}} \left\langle \phi^{\mathrm{S}}(\mathcal{D}^S), \mathcal{D}^T \right\rangle = \mathcal{A}\left(\phi^{\mathrm{S}}(\mathcal{D}^T) \right) \tag{10}$$

Phase 1: Baseline Transfer Learning At this stage, transfer learning is employed to adapt a pre-trained *distilbert-base-multilingual-cased* model, denoted as $\phi^{\mathrm{S}}(\cdot)$, to the target task, represented by $\phi^{\mathrm{T}}(\cdot)$. This adaptation is performed by applying a set of transformations $\mathcal{A}$ to its outputs, while keeping the original weights, $\mathcal{W}^S = 134{,}735{,}616$, frozen.

The feature extractor encodes input samples as tensors of dimension $\mathbb{R}^{N\times\lambda\times d}$. Subsequently, Global Max Pooling (GMP) reduces these tensors to $\mathbb{R}^{N\times d}$. The adaptation block consists of $\mathcal{W}^T = 104{,}294$ trainable parameters. It incorporates Batch Normalization (BN), a fully connected layer with 512 units (FC_{512}), Dropout with a probability of 0.1 ($\mathrm{Dro}_{0.1}$), two consecutive fully connected layers with 128 units each (FC_{128}), an additional Dropout layer, and an output layer with $\kappa = 6$ units (FC_κ), as detailed in Equations (7), (8), and (11).

$$\mathcal{A} = FC_6 \circ Dro_{0.1} \circ FC_{128} \circ Dro_{0.1} \circ FC_{128} \circ Dro_{0.1} \circ FC_{512} \circ BN \circ GMP \quad (11)$$

Due to hardware constraints, a reduced subset of the data was employed in this phase: $d^T_{\text{train}} : 20{,}000 \subset D^T_{\text{train}}$, $d^T_{\text{valid}} : 4{,}000 \subset D^T_{\text{valid}}$, and $d^T_{\text{test}} : 4{,}000 \subset D^T_{\text{test}}$. The training parameters utilized in this stage are summarized in Table 3 Phase 1.

Phase 2: Fine-Tuning on Dataset $D_{\kappa=6}$ In this phase, we adopt FT for all the pre-trained model's weights, as represented in (10), leveraging the inherent advantage that full FT does not necessitate supplementary output layers. As stated in [23], this procedure reduces loss and increases classification performance on the new task. Table 3 details the training hyperparameters utilized to FT the model in the basic taxonomy dataset in the Phase 2 column.

Phase 3: Scaling to Dataset $D_{\kappa=11}$ In this phase, we extend the scope of the FT model to the $D_{\kappa=11}$ dataset, which expands to a total of 11 categories. Procedures for building the aforementioned dataset were described in Section 3.1. The dataset was split into training, validation, and testing sets with 807,468, 201,867, and 100,000 samples, respectively. To FT the model using such a large dataset, we used a TPU v2[10]. The hyperparameters settings for this phase are presented in Phase 3 of Table 3.

4 Experimental Results

Tables 4, 5, and 6 report the model's performance across Phases 1, 2, and 3, respectively. FT the model on the extended dataset with TPU acceleration led to improved performance and enabled the prediction of a broader range of robbery categories. The observed discrepancies in support counts between Phase 1 and Phase 2 are attributed to changes in category labeling introduced by CEESJCT. It is also evident that the distribution of samples across categories is imbalanced, reflecting the underlying social circumstances surrounding criminal activity. Consequently, we opted not to employ synthetic data augmentation for

Table 3. Training Parameters Across Phases

Parameter	Phase 1 (TL)	Phase 2 (FT)	Phase 3 (FT-extended)
Loss	Categorical Cross Entropy	Sparse Categorical Cross Entropy	Sparse Categorical Cross Entropy
Metrics	Categorical Accuracy	Sparse Categorical Accuracy	Sparse Categorical Accuracy
Optimizer	Adam	Adam	Adam
Learning rate	2×10^{-4}	3×10^{-6}	3×10^{-6}
Epochs	20	10	12
Early stopping	5 epochs	10 epochs	10 epochs
Batch size	16	16	128

Table 4. Phase 1 TL Classifier Performance on $D_{\kappa=6}$

class	precision	recall	f1-score	support
Car Theft	0.58	0.36	0.45	285
Motorcycle Theft	0.90	0.84	0.87	457
Vehicle Parts And Accessories Theft	0.79	0.79	0.79	613
Street Robbery	0.82	0.91	0.86	1,666
Business Robbery	0.82	0.70	0.75	328
Home Robbery	0.76	0.77	0.77	651
micro avg	0.80	0.80	0.80	4,000
macro avg	0.78	0.73	0.75	4,000
weighted avg	0.80	0.80	0.80	4,000

Table 5. Phase 2 FT Classifier Performance on $D_{\kappa=6}$

class	precision	recall	f1-score	support
Business Robbery	0.78	0.79	0.78	297
Car Theft	0.95	0.88	0.92	363
Motorcycle Theft	0.95	0.93	0.94	392
Vehicle Parts & Accessories	0.85	0.92	0.88	663
Home Robbery	0.87	0.89	0.88	663
Street Robbery	0.94	0.92	0.93	1,671
Accuracy			0.90	4,000
Macro Avg	0.89	0.89	0.89	4,000
Weighted Avg	0.90	0.90	0.90	4,000

this study. The results further suggest that the *Public Institution Robbery* category should be merged into the *Other Robberies* category, due to the limited number of available samples.

5 Conclusions and Future Works

Transformer-based models have achieved state-of-the-art performance across a range of NLP tasks. In this work, we employed FT to address text classification for legal documents. The proposed methodology, based on FT, demonstrates that transformer architectures can be effectively adapted to the legal domain. Notably, our model demonstrates substantial robustness to class imbalance, with

Table 6. Phase 3 FT Classifier Performance on $D_{\kappa=11}$

	precision	recall	f1-score	support
Educational Institution Robbery	0.925995	0.946599	0.936184	1,573
Motorcycle Theft	0.985211	0.988317	0.986762	10,785
Public Institution Robbery	0.724528	0.474074	0.573134	405
Car Theft	0.967669	0.980198	0.973893	7,878
Social Organization Robbery	0.750000	0.752688	0.751342	279
Street Robbery	0.969847	0.972298	0.971071	37,976
Business Robbery	0.878871	0.893877	0.886311	6,794
Home Robbery	0.944779	0.952335	0.948542	15,504
Vehicle Parts & Accessories Theft	0.977752	0.968340	0.973023	14,024
Water Vessel Robbery	0.965197	0.983452	0.974239	846
Other Robberies	0.817738	0.766006	0.791027	3,936
accuracy			0.954610	100,000
macro avg	0.900690	0.879835	0.887775	100,000
weighted avg	0.954050	0.954610	0.954175	100,000

the exception of the *Public Institution Robbery* category. Due to the limited number of instances associated with this particular type of robbery, it may be advisable to consolidate it within the broader *Other Robberies* category. The utilization of TPU hardware was pivotal for processing the expanded dataset, enabling both an increase in the number of target categories and overall model performance. These results highlight the practical potential of deploying ML solutions within legal institutions such as the FGE, where automated narrative classification may facilitate more efficient resource allocation and optimized operational workflows. Future research should prioritize the adaptation of novel NLP tasks, as addressed by transformer-based models, to LNLP. Such tasks encompass, but are not limited to, legal question answering, legal named entity recognition, and legal judgment prediction. The latter, in particular, necessitates an increased emphasis on model explainability, given the importance of interpretability for legal stakeholders.

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References

1. Abnar, S., Zuidema, W.: Quantifying attention flow in transformers. Proceedings of the Annual Meeting of the Association for Computational Linguistics pp. 4190–4197 (5 2020). https://doi.org/10.18653/v1/2020.acl-main.385, https://arxiv.org/abs/2005.00928v2
2. Akca, O., Bayrak, G., Issifu, A.M., Ganiz, M.C.: Traditional machine learning and deep learning-based text classification for turkish law documents using transformers and domain adaptation. 16th International Conference on INnovations in Intelligent SysTems and Applications, INISTA 2022 (2022). https://doi.org/10.1109/INISTA55318.2022.9894051

3. Alom, M.Z., Taha, T.M., Yakopcic, C., Westberg, S., Sidike, P., Nasrin, M.S., Essen, B.C.V., Awwal, A.A.S., Asari, V.K.: The history began from alexnet: A comprehensive survey on deep learning approaches. CoRR **abs/1803.01164** (2018), http://arxiv.org/abs/1803.01164
4. Ariai, F., Demartini, G.: Natural language processing for the legal domain: A survey of tasks, datasets, models, and challenges. ACM Computing Surveys **1** (10 2024). https://doi.org/10.1145/nnnnnnn.nnnnnnn, https://arxiv.org/abs/2410.21306v2
5. Chalkidis, I., Fergadiotis, M., Androutsopoulos, I.: Legal-bert: The muppets straight out of law school. arXiv preprint arXiv:2010.02559 (2020)
6. Clavié, B., Alphonsus, M., Clavié, C., Mundi, J.: The unreasonable effectiveness of the baseline: Discussing svms in legal text classification (2021), https://gitlab.com/jusmundi-group/public/Legal-svm-baselines
7. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Burstein, J., Doran, C., Solorio, T. (eds.) Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). pp. 4171–4186. Association for Computational Linguistics, Minneapolis, Minnesota (Jun 2019). https://doi.org/10.18653/v1/N19-1423, https://aclanthology.org/N19-1423/
8. Falconí, L.G.: Transfer and ensemble learning models in breast mammogram pathology classification. Master's thesis, Quito, 2020. (2020)
9. Howard, J., Ruder, S.: Universal language model fine-tuning for text classification (2018), https://arxiv.org/abs/1801.06146
10. Jouppi, N.P., Yoon, D.H., Kurian, G., Li, S., Patil, N., Laudon, J., Young, C., Patterson, D.: A domain-specific supercomputer for training deep neural networks. Communications of the ACM **63**(7), 67–78 (2020)
11. Kalia, A., Kumar, N., Namdev, N.: Classifying case facts and predicting legal decisions of the indian central information commission: a natural language processing approach. In: Advances in Deep Learning, Artificial Intelligence and Robotics, pp. 35–45. Springer (2022)
12. Kowsari, K., Meimandi, K.J., Heidarysafa, M., Mendu, S., Barnes, L., Brown, D.: Text classification algorithms: A survey (2019). https://doi.org/10.3390/info10040150, www.mdpi.com/journal/information
13. Liu, Z., Li, Z., Zhang, H.: Legal knowledge-enhanced language models for legal text processing. ICAIL (2022)
14. Minaee, S., Kalchbrenner, N., Cambria, E., Nikzad, N., Chenaghlu, M., Gao, J.: Deep learning–based text classification. ACM Computing Surveys (CSUR) **54** (4 2021). https://doi.org/10.1145/3439726, https://dl.acm.org/doi/10.1145/3439726
15. Mumcuoğlu, E., Öztürk, C.E., Ozaktas, H.M., Koç, A.: Natural language processing in law: Prediction of outcomes in the higher courts of turkey. Information Processing & Management **58**(5), 102684 (2021)
16. Murphy, K.P.: Probabilistic machine learning: an introduction. MIT press (2022)
17. Pathak, A.R., Pandey, M., Rautaray, S.: Application of deep learning for object detection. Procedia Computer Science **132**, 1706–1717 (2018). https://doi.org/https://doi.org/10.1016/j.procs.2018.05.144, https://www.sciencedirect.com/science/article/pii/S1877050918308767, international Conference on Computational Intelligence and Data Science
18. Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al.: Language models are unsupervised multitask learners. OpenAI blog **1**(8), 9 (2019)

19. Sanh, V., Debut, L., Chaumond, J., Wolf, T.: Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. ArXiv **abs/1910.01108** (2019)
20. Savelka, J., Ashley, K.D.: Cross-domain generalization and knowledge transfer in transformers for legal text. ICAIL (2021)
21. Shaheen, Z., Wohlgenannt, G., Filtz, E.: Large scale legal text classification using transformer models (10 2020), https://arxiv.org/abs/2010.12871v1
22. Shaheen, Z., Wohlgenannt, G., Mouromtsev, D.: Zero-shot cross-lingual transfer in legal domain using transformer models. Proceedings - 2021 International Conference on Computational Science and Computational Intelligence, CSCI 2021 pp. 450–456 (11 2021). https://doi.org/10.1109/CSCI54926.2021.00145, https://arxiv.org/abs/2111.14192v2
23. Tunstall, L., von Werra, L., Wolf, T.: Natural language processing with transformers. " O'Reilly Media, Inc." (2022)
24. Vaswani, A., Shazeer, N., Parmar, N., et al.: Attention is all you need. Advances in Neural Information Processing Systems **30** (2017)
25. Vatsal, S., Meyers, A., Ortega, J.E.: Classification of us supreme court cases using bert-based techniques (2023), https://huggingface.co/saibo/
26. Wang, Y., Gao, J., Chen, J.: Deep learning algorithm for judicial judgment prediction based on bert. In: 2020 5th International Conference on Computing, Communication and Security (ICCCS). pp. 1–6. IEEE (2020)
27. Yan, G., Li, Y., Shen, S., Zhang, S., Liu, J.: Law article prediction based on deep learning. In: 2019 IEEE 19th International Conference on Software Quality, Reliability and Security Companion (QRS-C). pp. 281–284. IEEE (2019)
28. Yosinski, J., Clune, J., Bengio, Y., Lipson, H.: How transferable are features in deep neural networks? Advances in neural information processing systems **27** (2014)
29. Zhong, H., Xiao, C., Tu, C., Zhang, T., Liu, Z., Sun, M.: How does nlp benefit legal system: A summary of legal artificial intelligence. Proceedings of the Annual Meeting of the Association for Computational Linguistics pp. 5218–5230 (2020). https://doi.org/10.18653/V1/2020.ACL-MAIN.466, https://aclanthology.org/2020.acl-main.466/